

AI Support Agent for AppleSupport

Hiver SDE Intern Take-Home Assignment — Technical Report

Executive summary. I built an end-to-end AppleSupport support-agent prototype that classifies customer messages, retrieves similar historical resolutions, drafts a grounded response with an LLM, and decides whether to AUTO-HANDLE or ESCALATE. I first created a 200-example hand-labelled golden set, then scaled the classification experiment to 100K historical AppleSupport pairs using weak supervision. Multiple classical models and DistilBERT were evaluated on the same untouched 40-example golden test set. The larger weakly-supervised experiment reached approximately 50% accuracy, while the golden-only Logistic Regression baseline reached 37.5%. The results suggest that label quality and evaluation-set size are currently stronger bottlenecks than model capacity.

1. Problem Framing

For AppleSupport, a useful agent must (1) triage a message into an actionable intent, (2) draft a reply grounded in how similar historical conversations were resolved, and (3) recognize when automation is unsafe. I therefore optimized for **precision of automation over recall**: escalating a difficult message is preferable to confidently sending an incorrect answer.

The dataset contains noisy Twitter/X language, short or incomplete messages, slang, overlapping symptoms, and multi-topic requests. The prototype intentionally does not include multi-turn memory, production serving infrastructure, a learned retrieval reranker, active-learning label correction, or a separate production moderation layer.

2. Dataset and Conversation Construction

I selected **AppleSupport** from the Twitter Customer Support Conversations dataset. AppleSupport provides more than 100K usable customer-support pairs. I constructed pairs by linking AppleSupport response tweets to the original customer tweet through `in_response_to_tweet_id`, recovering the customer text, and removing pairs where the original customer message was unavailable.

Critical distinction: the historical ~106K conversations are not human-labelled intents. The trusted ground truth is the separate 200-example golden set. The 100K training experiment uses rule-based weak supervision.

3. Intent Taxonomy

#	Intent
1	Software / iOS Update
2	Device / Hardware Problem
3	Battery / Charging
4	App Problem
5	Apple Services
6	Connectivity
7	Messaging / Notifications
8	Account / Apple ID
9	Purchase / Activation
10	Payment / Billing
11	Complaint / Feedback
12	Other / Unclear

The taxonomy is compact and derived from observed AppleSupport patterns. It is intended to map to distinct support workflows, while acknowledging overlap between issues such as software updates, battery symptoms, and hardware problems.

4. Golden Set and Evaluation Design

Exactly **200 examples** were hand-labelled using a fixed random seed (`random_state=42`). The golden set was separated into approximately **160 training and 40 test examples**. The 40 test examples remain untouched for model comparison.

Intent	Count
Software / iOS Update	58
Other / Unclear	47

Intent	Count
Device / Hardware Problem	25
Battery / Charging	14
Connectivity	12
App Problem	9
Complaint / Feedback	9
Messaging / Notifications	9
Apple Services	7
Purchase / Activation	6
Account / Apple ID	3
Payment / Billing	1

Payment / Billing has one golden example, so it could not be placed in a stratified held-out test partition. It was retained in training; its test performance is unmeasured. With only 40 test examples, one prediction changes accuracy by 2.5 percentage points.

5. Classification Experiments

I first established simple baselines using the trusted golden training split. I then removed all golden examples from the historical pool, weak-labelled the remaining historical messages with high-precision rules, sampled 100K examples, and trained several model families. Every 100K experiment uses the same untouched 40-example test set.

Model	Training	Accuracy	Macro F1
Majority baseline	Golden (160)	30.0%	4.2%
TF-IDF + Logistic Regression	Golden (160)	37.5%	24.3%
Multinomial Naive Bayes	100K weak	35.0%	15.4%
Logistic Regression	100K weak	47.5%	47.5%
SGD Classifier	100K weak	50.0%	48.0%
Random Forest	100K weak	50.0%	48.0%
Ridge Classifier	100K weak	50.0%	48.1%
Linear SVM	100K weak	50.0%	49.8%
DistilBERT (fine-tuned)	100K weak	50.0%	49.8%

The larger weakly-supervised experiment improves over the golden-only baseline but plateaus around 50%. This suggests that **label quality is currently a stronger bottleneck than model choice**; it is not evidence of a theoretical 50% ceiling.

The end-to-end prototype uses the golden-only Logistic Regression classifier because the safety gate needs usable probability estimates through `predict_proba()`. This is an operational trade-off, not a claim that Logistic Regression is the most accurate classifier.

6. 100K Weak Supervision

Because the historical conversations have no intent labels, I generated labels with high-precision rules covering battery/charging, messaging, Apple ID, payment, connectivity, Apple services, software updates, app problems, hardware symptoms, and explicit complaints. Unmatched messages fall into Other / Unclear. Golden-set tweet IDs were excluded before selecting exactly 100K examples.

The advantage is scale; the cost is label noise. Multi-topic messages can be assigned according to a heuristic signal rather than the customer's primary need. Consequently, the 100K experiment should be interpreted as a weak-supervision study, not as training on 100K ground-truth labels.

7. Retrieval and Grounding

The response pipeline retrieves historical AppleSupport resolutions using Sentence Transformers **all-MiniLM-L6-v2** and FAISS with normalized embeddings. A 30K historical sample was indexed for practicality, with all golden examples excluded.

Pure semantic similarity sometimes retrieved a superficially similar but incorrect topic—for example, Wi-Fi/connectivity cases for an App Problem. I therefore filtered retrieval candidates by predicted intent before semantic ranking. This improved topical consistency but still inherits classifier and weak-label noise.

8. Grounded Response Generation

An LLM drafts a concise, professional customer-facing response from the retrieved evidence. The prompt discourages inventing refunds, policies, guarantees, technical claims, or unsupported URLs. The main observed weakness is **generic DM-deflection**: a customer asking a direct informational question can receive a request to move to DM because that pattern is common in the historical support corpus.

9. AUTO-HANDLE vs ESCALATE

Policy	AUTO-HANDLE	ESCALATE	Wrong AUTO-HANDLE
Initial/permissive	27/40 (67.5%)	13/40 (32.5%)	18/40 (45.0%)
Final conservative	1/40 (2.5%)	39/40 (97.5%)	0/40 (0.0%)

The final policy is intentionally conservative. The 0% wrong-auto-handle result is a safety-boundary result, not evidence of broad automation: only one of 40 cases was auto-handled. The permissive policy demonstrates the opposite trade-off—much higher coverage but unacceptable wrong automation.

10. LLM-as-Judge Evaluation

Generated replies were evaluated with a separate LLM judge, **Qwen/Qwen3.8-27B** through Hugging Face inference. Five criteria were scored from 0–2: relevance, groundedness, helpfulness, professionalism, and hallucination-free behavior.

Criterion	Average
Relevance	1.4 / 2
Groundedness	1.4 / 2
Helpfulness	0.9 / 2
Professionalism	1.9 / 2
Hallucination-free	1.9 / 2
Overall	7.5 / 10

Professionalism and hallucination avoidance are relatively strong; helpfulness is weaker. The audit contains only 10 examples, so the judge score is directional rather than a validated quality estimate.

11. Judge–Human Agreement

Criterion	Exact agreement	Agreement statistic
Relevance	60%	Quadratic weighted κ = 0.091
Groundedness	50%	κ = -0.250
Helpfulness	20%	κ = -0.038
Professionalism	50%	κ = -0.190
Hallucination-free	90%	κ = 0.000
Overall	—	MAE = 1.3 / 10

The judge aligns best with humans on hallucination-free behaviour and poorly on subjective dimensions such as helpfulness. With $n=10$, kappa is unstable and should not be interpreted as a definitive judge-quality estimate. The LLM judge is therefore an auxiliary signal, not ground truth.

12. Top 5 Failure Modes

1. Software-update vs hardware confusion. “the new Mac OS Sierra just messed my laptop up. Finder is not even working” → true Device / Hardware; predicted Software / Update. OS-version language dominates the heuristic/classifier.

2. Battery vs update confusion. “did the iOS 11.03 update and my battery still sucks” → true Battery / Charging; predicted Software / Update. Dual-topic messages are vulnerable to heuristic ordering.

3. Wrong retrieval despite semantic similarity. An App Problem message about apps not loading despite working Wi-Fi initially retrieved Wi-Fi/connectivity cases. Embeddings capture shared symptoms but not always the intended support workflow.

4. Generic DM-deflecting replies. “Any way to check battery health on my iPhone 6s?” received a generic DM request rather than a direct answer. The generator imitates a frequent historical resolution pattern.

5. Extremely conservative automation. The final safety policy auto-handled only 1/40 cases. Safety is good, but practical automation coverage is currently negligible.

13. What Is Misleading About My Headline Number?

The 50% accuracy figure is not production-grade evidence. It is measured on only 40 hand-labelled test examples, where each example is worth 2.5 percentage points. The 100K training labels are also pseudo-labels rather than ground truth. A model can learn the errors of the weak-supervision rules.

The 0% wrong-auto-handle figure is misleading without coverage: it comes from only one auto-handled example. The 7.5/10 LLM-judge result is based on only 10 replies and the judge-human audit shows weak agreement on several subjective criteria. Rare classes such as Payment/Billing and Account/Apple ID have too few golden examples for stable per-class conclusions.

Finally, the end-to-end prototype uses the golden-only Logistic Regression model, not the best 100K classifier. It was selected because its probability estimates support confidence-based safety gating. This means the strongest model-comparison score is not the same thing as the prototype’s operating classifier.

14. What I Would Do With One More Week

Day	Priority	Outcome
1	Tighten taxonomy and labeling guidelines	Resolve multi-topic cases by primary requested action.
2	Expand golden set to 500–1,000	Better class balance and stable evaluation.
3	Active learning for weak labels	Human-correct uncertain/high-impact pseudo-labels.
4	Re-train transformer	Test stronger models on cleaner labels.
5	Improve retrieval	Full 106K index + cross-encoder reranker.
6	Improve response safety	Direct-answer checks + grounding verification.
7	Larger end-to-end evaluation	Repeat judge/human audit at n=30–50.

15. Decision Log

#	Decision	Reason / trade-off
1	Select AppleSupport	Large single-brand history and recognizable patterns.
2	Use conversation pairs	Needed issue + historical resolution evidence.
3	Use 12 intents	Compact taxonomy maps to support workflows.
4	Hand-label 200 examples	Creates trusted evaluation ground truth.
5	Exclude golden IDs	Reduces direct evaluation leakage.
6	Keep single Payment/Billing example in training	Avoid invalid stratification; performance unmeasured.
7	Build golden-only baseline first	Establishes honest limited-label baseline.
8	Use weak supervision for 100K	Scales training but introduces label noise.
9	Compare model families	Tests whether model choice dominates data quality.
10	Fine-tune DistilBERT	Tests pretrained transformer under same noisy labels.
11	Use 30K retrieval sample	Practical embedding/indexing trade-off.
12	Use Logistic Regression in prototype	Probability estimates support safety gating.
13	Filter retrieval by predicted intent	Reduces wrong-topic semantic matches.
14	Use LLM for drafting	Avoids copying historical replies verbatim; adds generation risk.
15	Harden escalation policy	Reduced 45% wrong-auto-handle rate at cost of coverage.

16. Reproducibility and Submission Checklist

Core flow: customer message → intent prediction → confidence → historical retrieval → evidence → grounded LLM draft → safety policy → AUTO-HANDLE or ESCALATE.

Stack: Python, pandas, scikit-learn, Sentence Transformers, FAISS, Hugging Face Transformers, DistilBERT, Gemini for response generation, and Qwen3.8-27B for judging. Transformer training used a Tesla T4 GPU.

Requirement	Status
Runnable repository + README	Covered
150–250 hand-labelled golden set	Covered — 200
Sampling / labeling note	Covered
Automated metrics	Covered — accuracy, Macro F1, safety metrics
LLM-as-judge rubric	Covered
Judge-human agreement	Covered — n=10
Two baselines	Covered
Top 5 failure modes	Covered with real examples
Misleading headline number	Covered
One-week plan	Covered
10–15 decision log	Covered — 15 decisions

17. Conclusion

The prototype demonstrates the complete mechanics of an AI support agent rather than treating classification as the entire solution. It combines intent classification, historical grounding, LLM response generation, and explicit human escalation. The main measured limitation is clear: expanding from trusted labels to 100K weak labels improves the result but does not unlock strong accuracy, and several model families plus DistilBERT converge near 50% on the small golden test. The next highest-value work is therefore better labels, a larger balanced evaluation set, improved retrieval ranking, and direct-answer/safety checks—not simply trying more model architectures.